

Hematopoietic_Stem_Cell_Differentiation

June 1, 2026

1 Trajectory Analysis

One of the applications of SC RNA-Seq is the ability to understand how cells evolve in their transcriptional landscape over time, basically this is a trajectory of the transcription over time.

Trajectory analysis tries to order single cells along a continuous path that represents a biological process such as differentiation, activation, or response to stimulus. It does not give absolute time. Instead it gives pseudotime, a relative ordering that puts “earlier” cells at one end and “later” cells at the other.

Use trajectory analysis when: - You expect a continuous process (development, differentiation, activation). - Cells are sampled from different stages of that process, not just random unrelated cell types. - Do not use it when the population is a mix of distinct unrelated cell types with no transitional states.

In our case study, we are working with Fetal Liver hematopoietic stem cell that has the potential to differentiate into different types of blood cells

1.0.1 Install Packages!

```
[ ]: !pip install scanpy
      !pip install anndata
      !pip3 install igraph
      !pip install celltypist
      !pip install decoupler
      !pip install fa2-modified
      !pip install louvain
      !pip install scvelo
```

Collecting scanpy

Downloading scanpy-1.11.5-py3-none-any.whl.metadata (9.3 kB)

Collecting anndata>=0.8 (from scanpy)

Downloading anndata-0.12.4-py3-none-any.whl.metadata (9.9 kB)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)

Collecting legacy-api-wrap>=1.4.1 (from scanpy)

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 Requirement already satisfied: matplotlib>=3.7.5 in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)
 Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: networkx>=2.7.1 in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (3.5)
 Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)
 Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-
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 /usr/local/lib/python3.12/dist-packages (from scanpy) (25.0)
 Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-
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 /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.13)
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 Requirement already satisfied: seaborn>=0.13.2 in
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 Collecting session-info2 (from scanpy)
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 Requirement already satisfied: statsmodels>=0.14.5 in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)
 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
 (from scanpy) (4.67.1)
 Requirement already satisfied: typing-extensions in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)
 Requirement already satisfied: umap-learn>=0.5.6 in
 /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)
 Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)
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 Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.8->scanpy)
 Downloading zarr-3.1.3-py3-none-any.whl.metadata (10 kB)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)
 Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.7.5->scanpy) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
 (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy) (3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)

Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

 Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)

Collecting numcodecs>=0.14 (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

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Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)

Collecting crc32c>=2.7 (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)

 Downloading crc32c-2.8-cp312-cp312-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.8 kB)

Downloading scanpy-1.11.5-py3-none-any.whl (2.1 MB)

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Downloading anndata-0.12.4-py3-none-any.whl (170 kB)

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Downloading legacy_api_wrap-1.4.1-py3-none-any.whl (10.0 kB)

Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)

Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)

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Downloading zarr-3.1.3-py3-none-any.whl (276 kB)

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Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)

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numcodecs-0.16.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.8 MB)

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Downloading crc32c-2.8-cp312-cp312-

manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (80 kB)

80.0/80.0 kB

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Installing collected packages: session-info2, numcodecs, legacy-api-wrap, donfig, crc32c, array-api-compat, zarr, anndata, scanpy

Successfully installed anndata-0.12.4 array-api-compat-1.12.0 crc32c-2.8 donfig-0.8.1.post1 legacy-api-wrap-1.4.1 numcodecs-0.16.3 scanpy-1.11.5 session-info2-0.2.3 zarr-3.1.3

Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (0.12.4)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.12.0)

Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.15.1)

Requirement already satisfied: legacy-api-wrap in /usr/local/lib/python3.12/dist-packages (from anndata) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata) (8.4.0)

Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.0.2)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata) (25.0)

Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)

Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.16.2)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.1.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)

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Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (0.16.3)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata) (4.15.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata) (2.8)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

Collecting igraph

 Downloading igraph-1.0.0-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (4.4 kB)

Collecting texttable>=1.6.2 (from igraph)

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 Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)

Installing collected packages: texttable, igraph

Successfully installed igraph-1.0.0 texttable-1.7.0

Collecting celltypist

 Downloading celltypist-1.7.1-py3-none-any.whl.metadata (45 kB)

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Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)

Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.2.2)

Requirement already satisfied: scikit-learn>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)

Requirement already satisfied: openpyxl>=3.0.4 in /usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)

Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from celltypist) (8.3.0)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)

Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.11.5)

Collecting leidenalg>=0.9.0 (from celltypist)

 Downloading leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (10 kB)

 Collecting igraph<0.12,>=0.10.0 (from leidenalg>=0.9.0->celltypist)

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Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl>=3.0.4->celltypist) (2.0.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celltypist) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celttypist) (2025.2)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celttypist) (2025.10.5)

Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.12.4)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (3.15.1)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (1.5.2)

Requirement already satisfied: legacy-api-wrap>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (1.4.1)

Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (3.5)

Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.60.0)

Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (25.0)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.5.13)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celttypist) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.15.0)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.5.9.post2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist) (3.6.0)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)

Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages

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(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8-
>scanpy>=1.7.0->celltypist) (2.8)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
7.3/7.3 MB
82.0 MB/s eta 0:00:00
Downloading
leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0
MB)
2.0/2.0 MB
93.7 MB/s eta 0:00:00
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igraph-0.11.9-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.4 MB)
4.4/4.4 MB
100.2 MB/s eta 0:00:00
Installing collected packages: igraph, leidenalg, celltypist
  Attempting uninstall: igraph
    Found existing installation: igraph 1.0.0
    Uninstalling igraph-1.0.0:
      Successfully uninstalled igraph-1.0.0
Successfully installed celltypist-1.7.1 igraph-0.11.9 leidenalg-0.10.2
Collecting decoupler
  Downloading decoupler-2.1.1-py3-none-any.whl.metadata (8.6 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.12.4)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) ... done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)
62.0/62.0 kB
5.2 MB/s eta 0:00:00
Requirement already satisfied: session-info2 in
/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages

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(from decoupler) (4.67.1)
Requirement already satisfied: numpy<2.5,>=1.23.5 in
/usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from adjusttext->decoupler) (3.10.0)
Requirement already satisfied: array-api-compat>=1.7.1 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (3.15.1)
Requirement already satisfied: legacy-api-wrap in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.4.1)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
packages (from anndata->decoupler) (8.4.0)
Requirement already satisfied: packaging>=24.2 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in
/usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.3)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from docrep->decoupler) (1.17.0)
Collecting legendkit (from marsilea->decoupler)
  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (4.5.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-
packages (from marsilea->decoupler) (0.13.2)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->decoupler) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.10.5)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.3.3)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(1.4.9)

```

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.3)

Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)

Requirement already satisfied: pyarrow>=10.0.1 in /usr/local/lib/python3.12/dist-packages (from pandas[parquet]->marsilea->decoupler) (18.1.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)

Downloading decoupler-2.1.1-py3-none-any.whl (117 kB)
118.0/118.0 kB

7.4 MB/s eta 0:00:00

Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3 MB)
37.3/37.3 MB

15.8 MB/s eta 0:00:00

Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)

Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
85.0/85.0 kB

9.5 MB/s eta 0:00:00

Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)

Building wheels for collected packages: docrep

Building wheel for docrep (setup.py) ... done

Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876 sha256=d312e1320e4dd34391dc3a6ed7a7edcf98c6d5b59a2bb1b8bebfad954a8186b9

Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca b57ce52da3e6fab7614836bd8c

Successfully built docrep

```

Installing collected packages: scipy, docrep, legendkit, adjusttext, marsilea,
decoupler
  Attempting uninstall: scipy
    Found existing installation: scipy 1.16.2
    Uninstalling scipy-1.16.2:
      Successfully uninstalled scipy-1.16.2
Successfully installed adjusttext-1.3.0 decoupler-2.1.1 docrep-0.3.2
legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3
Collecting fa2-modified
  Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (9.3
kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.1)
Downloading fa2_modified-0.4-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (620 kB)
620.8/620.8 kB
14.8 MB/s eta 0:00:00
Installing collected packages: fa2-modified
Successfully installed fa2-modified-0.4
Collecting louvain
  Downloading louvain-0.8.2.tar.gz (4.2 MB)
4.2/4.2 MB
64.0 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: igraph<0.12,>=0.10.0 in
/usr/local/lib/python3.12/dist-packages (from louvain) (0.11.9)
Requirement already satisfied: texttable>=1.6.2 in
/usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
(1.7.0)
Building wheels for collected packages: louvain
  Building wheel for louvain (setup.py) ... done
  Created wheel for louvain: filename=louvain-0.8.2-cp312-cp312-linux_x86_64.whl
size=971851
sha256=c07366ab80fe2c313e13549fe337cb8bfee1373c8d44f599cf688b4a113f5144
  Stored in directory: /root/.cache/pip/wheels/40/de/2b/bb7ed19d84727f9f299f20cd
34c42bba9c8bef7d83d2255c86
Successfully built louvain
Installing collected packages: louvain
Successfully installed louvain-0.8.2
Collecting scvelo
  Downloading scvelo-0.3.3-py3-none-any.whl.metadata (8.0 kB)
Requirement already satisfied: anndata>=0.7.5 in /usr/local/lib/python3.12/dist-
packages (from scvelo) (0.12.4)

```

Requirement already satisfied: scanpy>=1.5 in /usr/local/lib/python3.12/dist-packages (from scvelo) (1.11.5)

Collecting loompy>=2.0.12 (from scvelo)

 Downloading loompy-3.0.8.tar.gz (49 kB)

 49.6/49.6 kB

3.6 MB/s eta 0:00:00

 Preparing metadata (setup.py) ... done

Requirement already satisfied: umap-learn>=0.3.10 in /usr/local/lib/python3.12/dist-packages (from scvelo) (0.5.9.post2)

Requirement already satisfied: numba>=0.41.0 in /usr/local/lib/python3.12/dist-packages (from scvelo) (0.60.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from scvelo) (2.0.2)

Requirement already satisfied: pandas!=1.4.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from scvelo) (2.2.2)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scvelo) (1.15.3)

Requirement already satisfied: scikit-learn>=0.21.2 in /usr/local/lib/python3.12/dist-packages (from scvelo) (1.6.1)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from scvelo) (3.10.0)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (1.12.0)

Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (3.15.1)

Requirement already satisfied: legacy-api-wrap in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (8.4.0)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (25.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.7.5->scvelo) (3.1.3)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from loompy>=2.0.12->scvelo) (75.2.0)

Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from loompy>=2.0.12->scvelo) (8.3.0)

Collecting numpy-groupies (from loompy>=2.0.12->scvelo)

 Downloading numpy_groupies-0.11.3-py3-none-any.whl.metadata (18 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->scvelo) (2.9.0.post0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.41.0->scvelo) (0.43.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,>=1.1.1->scvelo) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=1.4.0,>=1.1.1->scvelo) (2025.2)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (1.5.2)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (3.5)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (0.5.13)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (0.2.3)

Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (0.14.5)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (4.67.1)

Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.5->scvelo) (4.15.0)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.21.2->scvelo) (3.6.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->scvelo) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.7.5->scvelo) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.7.5->scvelo) (0.16.3)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.7.5->scvelo) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.7.5->scvelo) (2.8)

Downloading scvelo-0.3.3-py3-none-any.whl (196 kB)

196.8/196.8 kB

12.2 MB/s eta 0:00:00

Downloading numpy_groupies-0.11.3-py3-none-any.whl (40 kB)

40.8/40.8 kB

3.9 MB/s eta 0:00:00

Building wheels for collected packages: loompy

Building wheel for loompy (setup.py) ... done

Created wheel for loompy: filename=loompy-3.0.8-py3-none-any.whl size=54012
sha256=3181bb398a316a51f130c7a5ee5c971a553cbb9ff94a51643f3e017bbcdfa3ec

Stored in directory: /root/.cache/pip/wheels/2e/45/28/2c7afc7f9eab47612e1b5090
cfb9ba0f00777625064039a6af

Successfully built loompy

Installing collected packages: numpy-groupies, loompy, scvelo

Successfully installed loompy-3.0.8 numpy-groupies-0.11.3 scvelo-0.3.3

1.0.2 Load Data

```
[ ]: #Import core single cell datasets
```

```
import scanpy as sc
import anndata as ad
import numpy as np
#import scvelo as scv
```

```
[ ]: !mkdir -p GSM5115832
!wget https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/
↳GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Fbarcodes.tsv.gz -O /content/GSM5115832/
↳barcodes.tsv.gz
!wget https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/
↳GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Ffeatures.tsv.gz -O /content/GSM5115832/
↳features.tsv.gz
!wget https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/
↳GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Fmatrix.mtx.gz -O /content/GSM5115832/
↳matrix.mtx.gz
```

--2025-10-29 05:37:58-- https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Fbarcodes.tsv.gz

Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)... 130.14.250.13,
130.14.250.31, 130.14.250.7, ...

Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.13|:443...
connected.

HTTP request sent, awaiting response... 200 OK

Length: 61513 (60K) [application/x-gzip]

Saving to: '/content/GSM5115832/barcodes.tsv.gz'

/content/GSM5115832 100%[=====>] 60.07K --.-KB/s in 0.1s

2025-10-29 05:37:59 (488 KB/s) - '/content/GSM5115832/barcodes.tsv.gz' saved

[61513/61513]

```
--2025-10-29 05:37:59-- https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Ffeatures.tsv.gz
Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)... 130.14.250.13,
130.14.250.31, 130.14.250.7, ...
Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.13|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 279361 (273K) [application/x-gzip]
Saving to: '/content/GSM5115832/features.tsv.gz'
```

```
/content/GSM5115832 100%[=====>] 272.81K 1.06MB/s in 0.3s
```

```
2025-10-29 05:37:59 (1.06 MB/s) - '/content/GSM5115832/features.tsv.gz' saved
[279361/279361]
```

```
--2025-10-29 05:37:59-- https://ftp.ncbi.nlm.nih.gov/geo/samples/GSM5115nnn/GSM5115832/suppl/GSM5115832%5FE11%5F5%5FmAGM%5FFL%5Fmatrix.mtx.gz
Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)... 130.14.250.13,
130.14.250.31, 130.14.250.7, ...
Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.13|:443...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 144967517 (138M) [application/x-gzip]
Saving to: '/content/GSM5115832/matrix.mtx.gz'
```

```
/content/GSM5115832 100%[=====>] 138.25M 25.5MB/s in 6.1s
```

```
2025-10-29 05:38:06 (22.6 MB/s) - '/content/GSM5115832/matrix.mtx.gz' saved
[144967517/144967517]
```

Note that this is another way to import data obtained from 10x.

```
[ ]: esc_adata = sc.read_10x_mtx('/content/GSM5115832/')
esc_adata.var_names_make_unique()
```

1.0.3 QC

```
[ ]: esc_adata.shape
```

```
[ ]: (12289, 31053)
```

```
[ ]: esc_adata.var.head()
```

```
[ ]:
      gene_ids      feature_types
Xkr4      ENSMUSG000000051951  Gene Expression
Gm1992     ENSMUSG000000089699  Gene Expression
Gm37381    ENSMUSG000000102343  Gene Expression
Rp1        ENSMUSG000000025900  Gene Expression
Sox17      ENSMUSG000000025902  Gene Expression
```

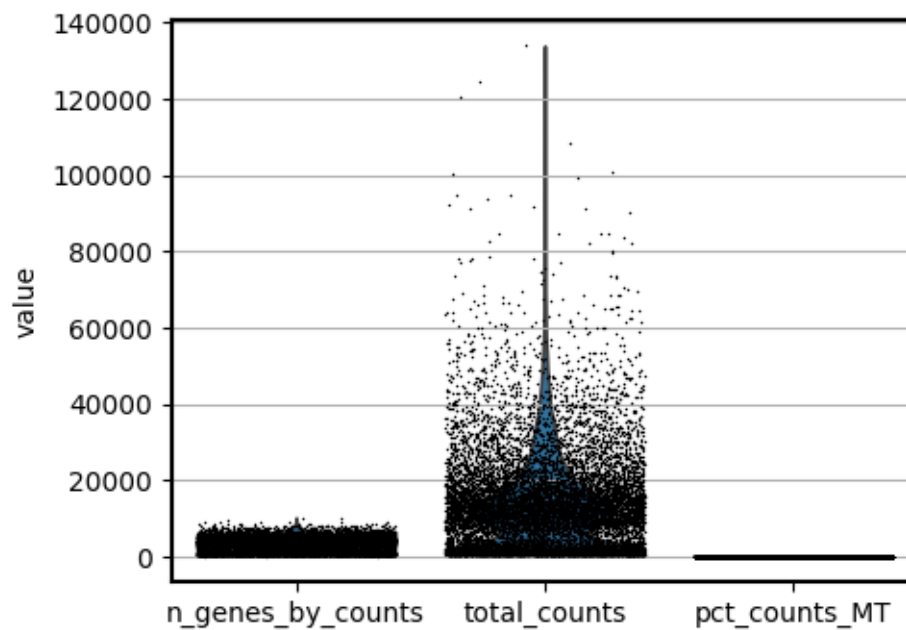
```
[ ]: esc_adata.var['MT'] = esc_adata.var_names.str.startswith("MT-")
      esc_adata.var['RIBO'] = esc_adata.var_names.str.startswith("RPS", "RPL")
      esc_adata.var['HB'] = esc_adata.var_names.str.startswith("^HB[^P]")
```

```
[ ]: sc.pp.calculate_qc_metrics(
      esc_adata, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)
```

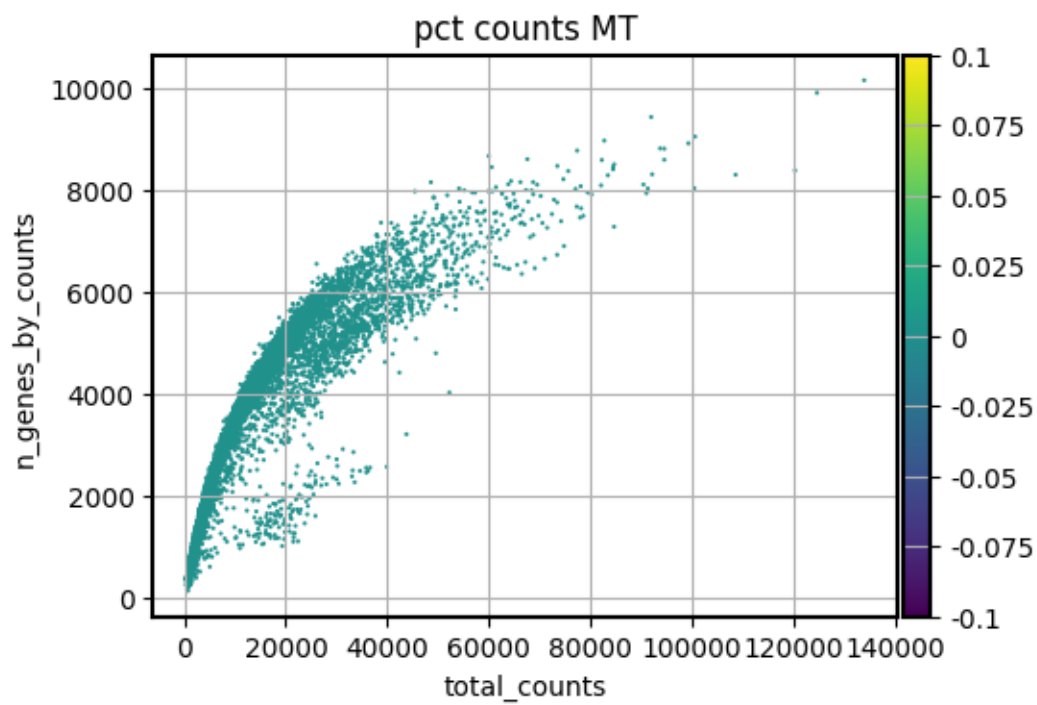
```
[ ]: import matplotlib.pyplot as plt

      plt.rcParams["figure.figsize"] = (5,4) # Adjust figure size
      plt.rcParams["axes.grid"] = True # Add grid to plots
      plt.rcParams["axes.edgecolor"] = "black" # Set plot border color
      plt.rcParams["axes.linewidth"] = 1.5 # Set plot border width
      plt.rcParams["axes.facecolor"] = "white" # Set background color
      plt.rcParams["axes.labelcolor"] = "black" # Set label color
      plt.rcParams["xtick.color"] = "black" # Set x-axis tick color
      plt.rcParams["ytick.color"] = "black" # Set y-axis tick color
      plt.rcParams["text.color"] = "black" # Set text color
```

```
[ ]: sc.pl.violin(
      esc_adata,
      ["n_genes_by_counts", 'total_counts', 'pct_counts_MT'],
      jitter=0.4,
      multi_panel=False,
)
```

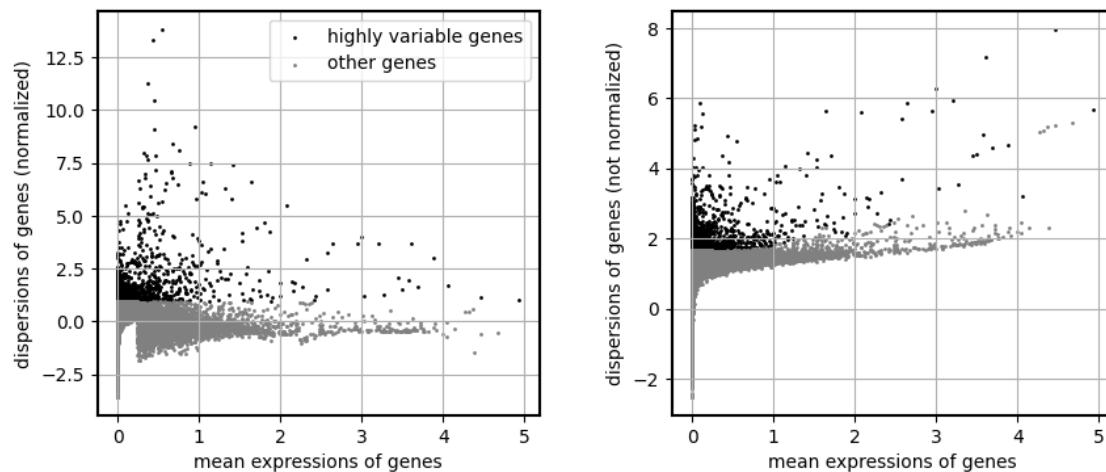
```
[ ]: sc.pl.scatter(esc_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_MT")
```



```
[ ]: #sc.pp.scrublet(esc_adata)
```

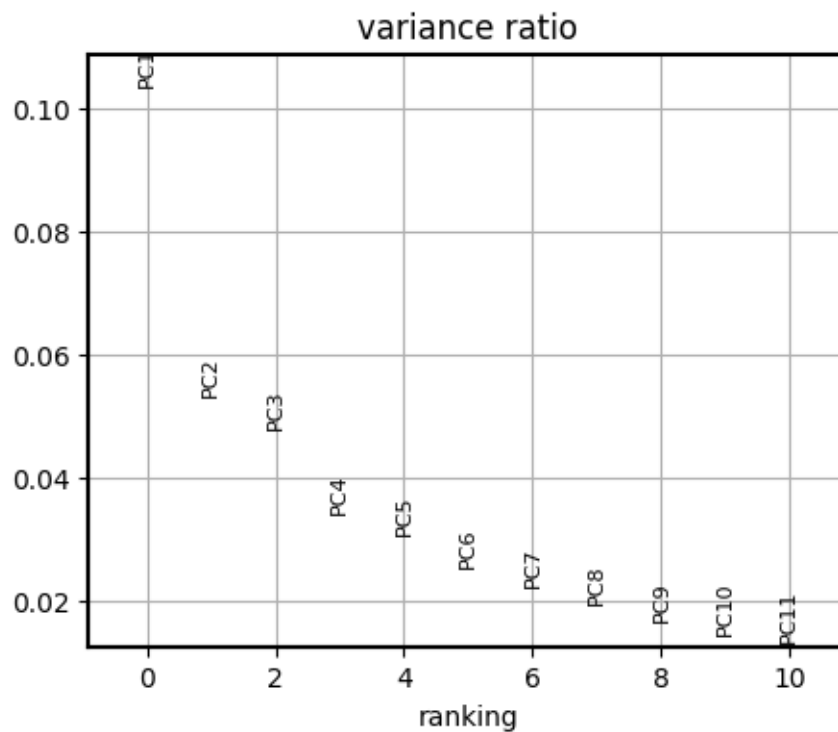
```
[ ]: #Normalisation  
esc_adata.layers["counts"] = esc_adata.X.copy()  
sc.pp.normalize_total(esc_adata)  
sc.pp.log1p(esc_adata)
```

```
[ ]: #Feature selection  
sc.pp.highly_variable_genes(esc_adata, n_top_genes=2500)  
sc.pl.highly_variable_genes(esc_adata)
```



1.0.4 Dimensionality Reduction

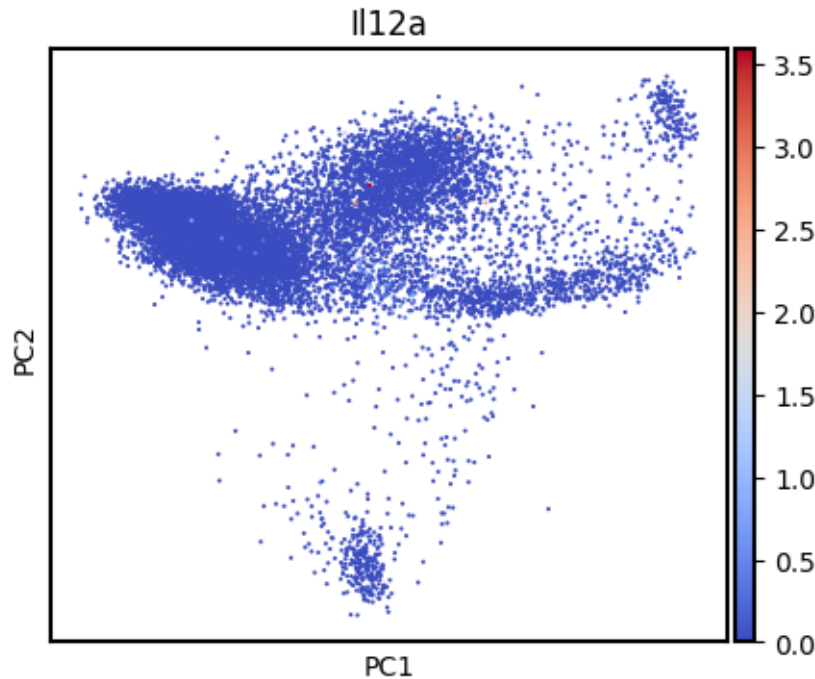
```
[ ]: #Dim Reduction  
sc.tl.pca(esc_adata)  
sc.pl.pca_variance_ratio(esc_adata, n_pcs=10, log=False)
```



```
[ ]: il_genes = [gene for gene in esc_adata.var_names[esc_adata.
    ↪var['highly_variable']] if gene.lower().startswith('il')]
print(il_genes)
```

```
['Il11b', 'Il2rg', 'Il13ra2', 'Il7', 'Il12a', 'Il6ra', 'Il17re', 'Il11', 'Il4i1',
'I115', 'Il2rb', 'Il33']
```

```
[ ]: sc.pl.pca(esc_adata, color="Il12a", cmap="coolwarm")
```



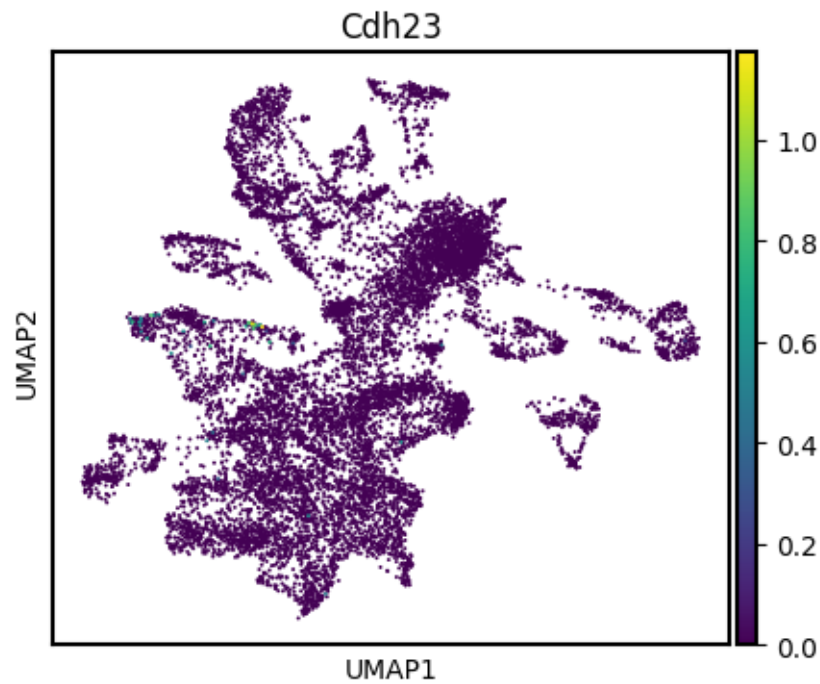
1.0.5 UMAP

```
[ ]: sc.pp.neighbors(esc_adata)
     sc.tl.umap(esc_adata)
```

```
[ ]: esc_adata
```

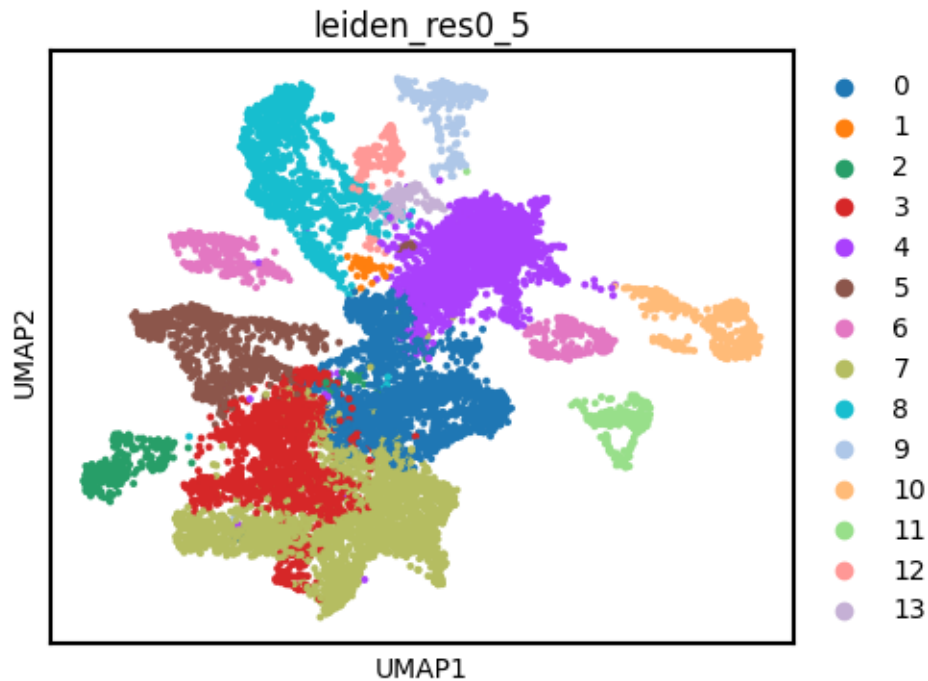
```
[ ]: AnnData object with n_obs × n_vars = 12289 × 31053
      obs: 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts',
          'log1p_total_counts', 'pct_counts_in_top_50_genes',
          'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
          'pct_counts_in_top_500_genes', 'total_counts_MT', 'log1p_total_counts_MT',
          'pct_counts_MT', 'total_counts_RIBO', 'log1p_total_counts_RIBO',
          'pct_counts_RIBO', 'total_counts_HB', 'log1p_total_counts_HB', 'pct_counts_HB'
      var: 'gene_ids', 'feature_types', 'MT', 'RIBO', 'HB', 'n_cells_by_counts',
          'mean_counts', 'log1p_mean_counts', 'pct_dropout_by_counts', 'total_counts',
          'log1p_total_counts', 'highly_variable', 'means', 'dispersions',
          'dispersions_norm'
      uns: 'log1p', 'hvg', 'pca', 'neighbors', 'umap'
      obsm: 'X_pca', 'X_umap'
      varm: 'PCs'
      layers: 'counts'
      obsp: 'distances', 'connectivities'
```

```
[ ]: sc.pl.umap(
    esc_adata,
    color=["Cdh23"],
    size=8,
)
```



```
[ ]: sc.tl.leiden(esc_adata, flavor="igraph", n_iterations=2,
    ↪key_added="leiden_res0_5", resolution=0.25)
```

```
[ ]: sc.pl.umap(
    esc_adata,
    color=["leiden_res0_5"],
    size=32,
)
```



1.0.6 Cell Type Annotation

```
[ ]: import decoupler as dc

[ ]: # Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="mouse")

# Keep canonical cell type markers alone
markers = markers[markers["mouse"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

# Format because dc only accepts cell_type and genesymbol
markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()
```

```
[ ]:      source  target
0   Hepatocytes   A1cf
1  Bergmann glia   A2m
2  Bergmann glia   Mug2
3  Bergmann glia   Mug1
4    Mast cells  A4galt
```

```
[ ]: esc_adata.var_names
```

```
[ ]: Index(['Xkr4', 'Gm1992', 'Gm37381', 'Rp1', 'Sox17', 'Gm37323', 'Mrpl15',
          'Lypla1', 'Gm37988', 'Tcea1',
          ...
          'AC125149.1', 'AC125149.2', 'AC125149.4', 'AC234645.1', 'AC168977.2',
          'AC168977.1', 'AC149090.1', 'CAAA01118383.1', 'Vmn2r122',
          'CAAA01147332.1'],
          dtype='object', length=31053)
```

```
[ ]: dc.mt.ulm(data=esc_adata,
              net=markers,
              tmin = 3)
```

```
[ ]: score = dc.pp.get_obs(esc_adata, key="score_ulm")
```

```
[ ]: esc_adata.obs["score_ulm"].head(1)
```

```
[ ]:      Acinar cells  Adipocyte progenitor cells  Adipocytes \
AAACCCACACAAAGCG-1      -0.750407                -0.551503   -0.617239

      Adrenergic neurons  Airway epithelial cells \
AAACCCACACAAAGCG-1      4.809056                -0.246552

      Airway goblet cells  Airway smooth muscle cells \
AAACCCACACAAAGCG-1      -0.59146                -0.246552

      Alpha cells  Alveolar macrophages \
AAACCCACACAAAGCG-1      -0.836877                -0.640888

      Anterior pituitary gland cells  ...  Thymocytes \
AAACCCACACAAAGCG-1                -0.652663  ...   -0.537526

      Transient cells  Trichocytes  Trigeminal neurons \
AAACCCACACAAAGCG-1      0.929286      -0.27566        -0.827713

      Trophoblast cells  Trophoblast progenitor cells \
AAACCCACACAAAGCG-1      -0.616667                -0.213516

      Tuft cells  Undefined placental cells  Urothelial cells \
```

```
AAACCCACACAAAGCG-1      0.53178      -0.427116      -0.444567
```

Vascular smooth muscle cells

```
AAACCCACACAAAGCG-1      -0.301977
```

```
[1 rows x 174 columns]
```

```
[ ]: esc_adata.obsm["score_ulm"].columns
```

```
[ ]: Index(['Acinar cells', 'Adipocyte progenitor cells', 'Adipocytes',
          'Adrenergic neurons', 'Airway epithelial cells', 'Airway goblet cells',
          'Airway smooth muscle cells', 'Alpha cells', 'Alveolar macrophages',
          'Anterior pituitary gland cells',
          ...,
          'Thymocytes', 'Transient cells', 'Trichocytes', 'Trigeminal neurons',
          'Trophoblast cells', 'Trophoblast progenitor cells', 'Tuft cells',
          'Undefined placental cells', 'Urothelial cells',
          'Vascular smooth muscle cells'],
          dtype='object', length=174)
```

```
[ ]: #rank genes
esc_adata_gene_rank = dc.tl.rankby_group(score, groupby="leiden_res0_5",
    ↪reference="rest", method="t-test_overestim_var")
esc_adata_gene_rank = esc_adata_gene_rank[esc_adata_gene_rank["stat"] > 0]
esc_adata_gene_rank.head(5)
```

```
[ ]:  group reference      name      stat  meanchange \
1      0      rest      Mesothelial cells  26.504235    0.709973
2      0      rest  Adipocyte progenitor cells  24.804045    0.869295
3      0      rest  Kidney progenitor cells  24.229891    0.580452
4      0      rest      Crypt cells  23.759255    0.484602
5      0      rest  Radial glia cells  23.309538    0.515700
```

```
      pval      padj
1  9.229752e-143  8.029885e-141
2  1.722461e-126  9.990271e-125
3  3.560492e-121  1.548814e-119
4  2.922803e-116  1.017135e-114
5  1.445739e-112  4.192644e-111
```

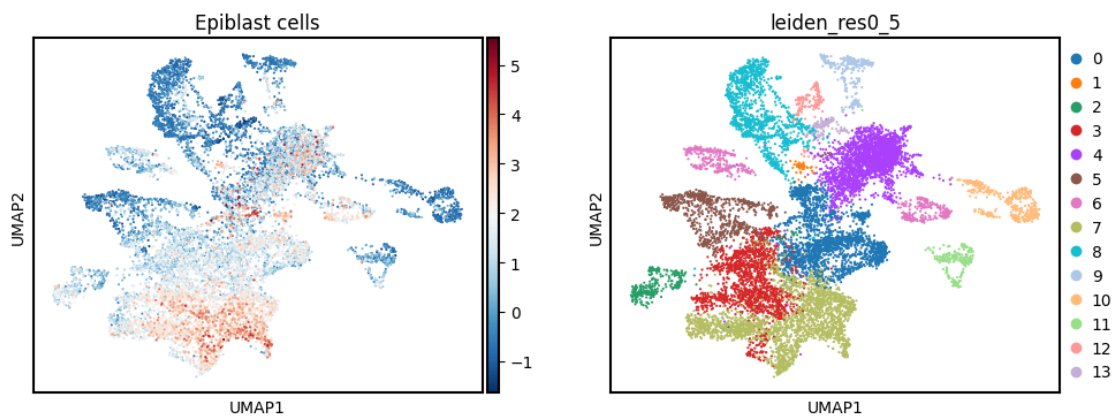
```
[ ]: top_cell_type_per_group = esc_adata_gene_rank.groupby('group')['name'].
    ↪apply(lambda x: x.head(1))
display(top_cell_type_per_group.to_dict())
```

```
{('0', 1): 'Mesothelial cells',
 ('1', 2262): 'Pluripotent stem cells',
 ('10', 1218): 'Hepatoblasts',
```



```
( '11', 1392): 'Endothelial cells',
( '12', 1740): 'Megakaryocytes',
( '13', 1914): 'Dendritic cells',
( '2', 2088): 'Anterior pituitary gland cells',
( '3', 348): 'Gamma delta T cells',
( '4', 174): 'T cells naive',
( '5', 1566): 'Sertoli cells',
( '6', 523): 'Glomus cells',
( '7', 696): 'Fibroblasts',
( '8', 870): 'Erythroblasts',
( '9', 1044): 'Erythroid-like and erythroid precursor cells'}
```

```
[ ]: sc.pl.umap(score, color=["Hepatoblasts","leiden_res0_5"], cmap="RdBu_r")
```

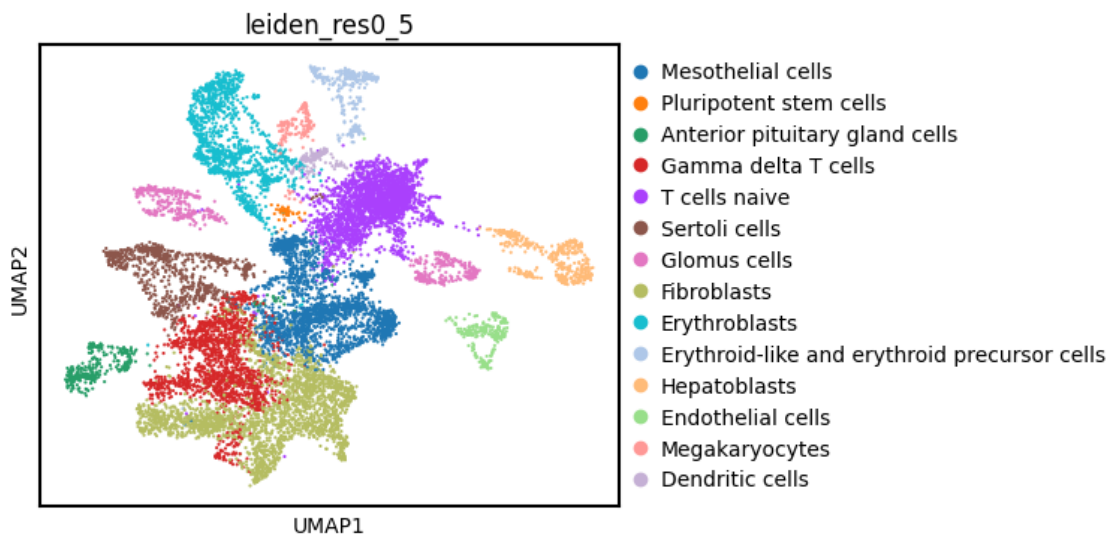


```
[ ]: dict_ann = esc_adata_gene_rank[esc_adata_gene_rank["stat"] > 0].
      ↳groupby("group").head(1).set_index("group")["name"].to_dict()
dict_ann
```

```
[ ]: { '0': 'Mesothelial cells',
      '4': 'T cells naive',
      '3': 'Gamma delta T cells',
      '6': 'Glomus cells',
      '7': 'Fibroblasts',
      '8': 'Erythroblasts',
      '9': 'Erythroid-like and erythroid precursor cells',
      '10': 'Hepatoblasts',
      '11': 'Endothelial cells',
      '5': 'Sertoli cells',
      '12': 'Megakaryocytes',
      '13': 'Dendritic cells',
      '2': 'Anterior pituitary gland cells',
      '1': 'Pluripotent stem cells'}
```

```
[ ]: esc_adata.obs["leiden_res0_5"] = esc_adata.obs["leiden_res0_5"].cat.  
      ↪ rename_categories(dict_ann)
```

```
[ ]: sc.pl.umap(  
      adata=esc_adata,  
      color=["leiden_res0_5"],  
      ncols=1,  
      )
```



1.0.7 Trajectory Inference

It is an attempt to understand how cells transition from one type to another (like stem → mature)?

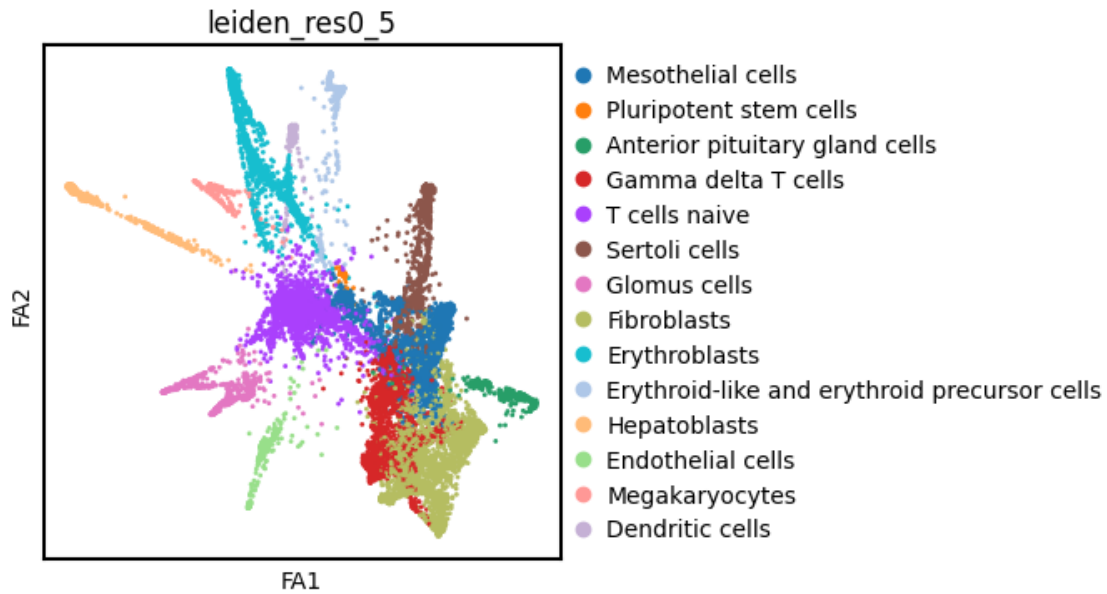
First, we will build a graph? Think of a **graph** as a network:

- Each **cell** is a **dot (node)**.
- If two cells are similar, draw a **line (edge)** between them.

So the graph connects cells that look alike.

```
[ ]: #Trajectory analysis  
sc.tl.draw_graph(esc_adata)
```

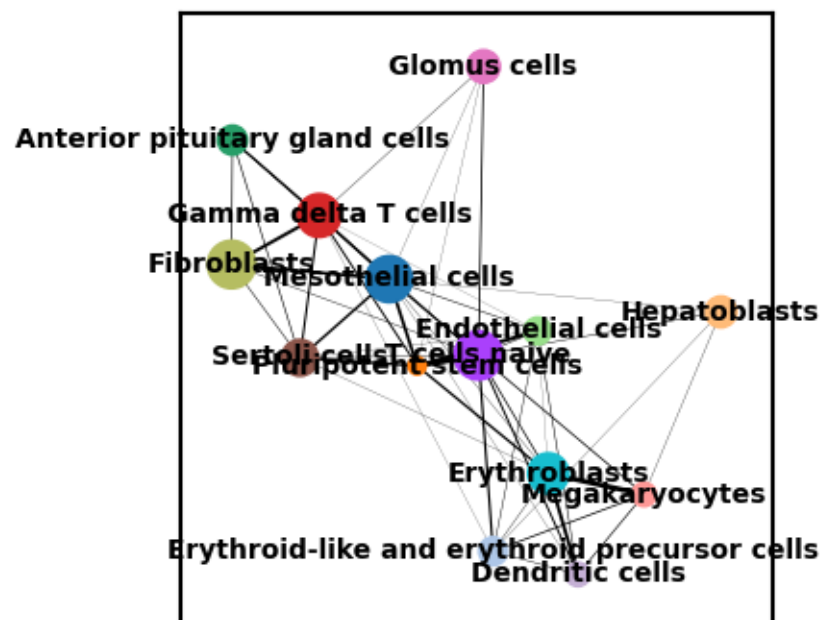
```
[ ]: plt.rcParams["figure.figsize"] = (4,4)  
sc.pl.draw_graph(esc_adata, color='leiden_res0_5', size = 16)
```



Then we will **ABSTRACT** the graph Basically, all the points that cluster to make one cell type, will be converted to one point. More like a blunt summary of every point

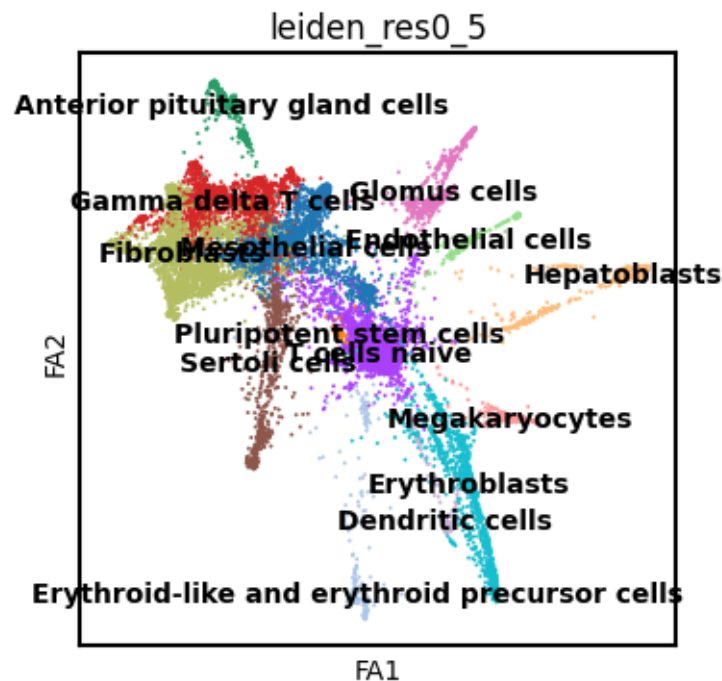
```
[ ]: sc.tl.paga(esc_adata, groups='leiden_res0_5')
```

```
[ ]: sc.pl.paga(esc_adata, color=['leiden_res0_5'])
```

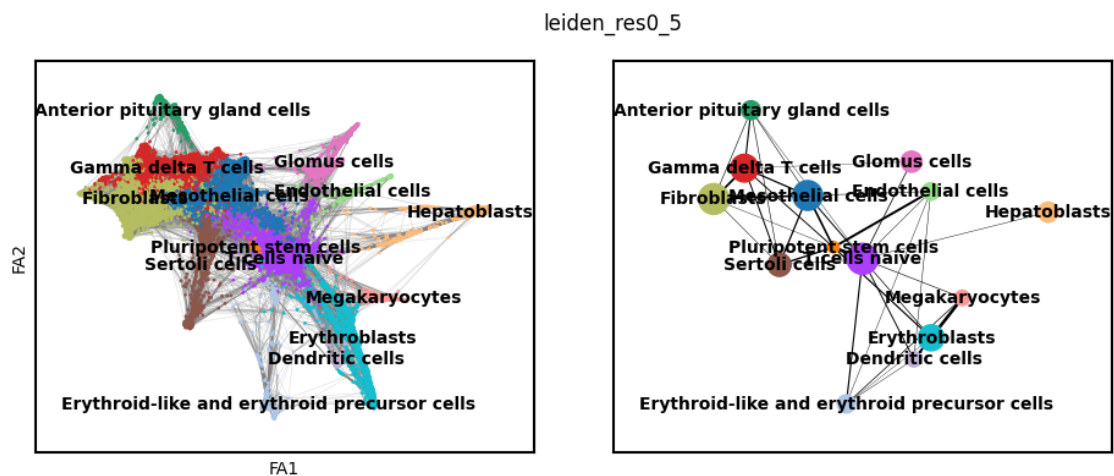


```
[ ]: sc.tl.draw_graph(esc_adata, init_pos='paga')
```

```
[ ]: sc.pl.draw_graph(esc_adata, color='leiden_res0_5', legend_loc='on data', size=8)
```



```
[ ]: plt.rcParams["figure.figsize"] = (5,4)
sc.pl.paga_compare(esc_adata, threshold=0.03, frameon=True, edges=True, size = 16)
```



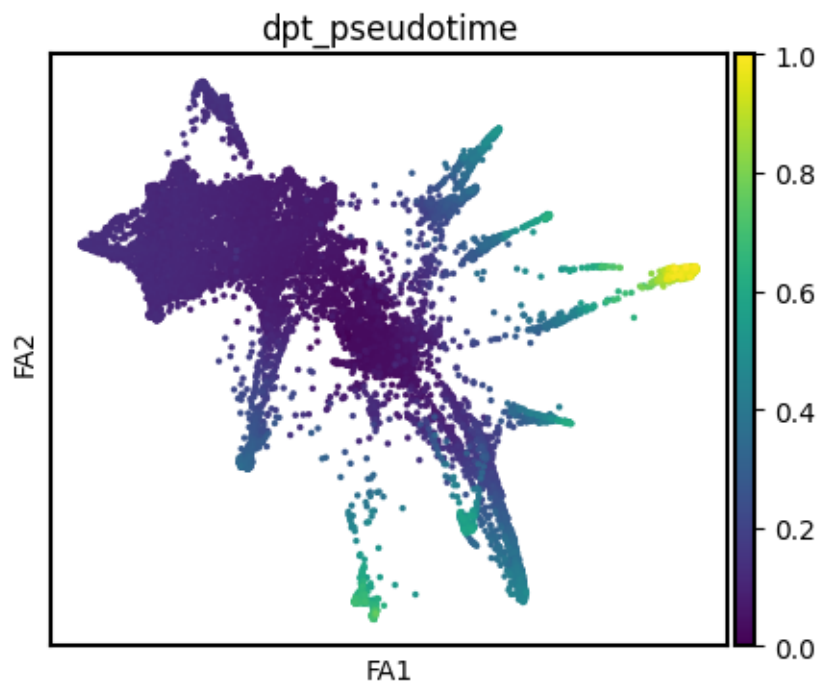
```
[ ]: [<Axes: xlabel='FA1', ylabel='FA2'>, <Axes: >]
```

Now how do cells transition from one type to another, assuming we have a pluripotent progenitor cell.

```
[ ]: esc_adata.uns['iroot'] = np.flatnonzero(esc_adata.obs['leiden_res0_5'] ==  
↳ 'Pluripotent stem cells')[0]  
sc.tl.dpt(esc_adata)
```

WARNING: Trying to run `tl.dpt` without prior call of `tl.diffmap`. Falling back to `tl.diffmap` with default parameters.

```
[ ]: sc.pl.draw_graph(esc_adata, color=['dpt_pseudotime', 'leiden_res0_5'],  
↳ legend_loc='on data', size = 24)
```



```
[ ]: esc_adata.write("esc_adata.h5", compression="gzip")
```